Four Foundations Facing the Same Mathematical Problems

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A theory of sets is tested by the constructions it permits. ZFC, von Neumann–Bernays–Goedel class theory (NBG), Kelley–Morse class theory (KM), and Quine's New Foundations (NF) differ most clearly when we ask whether the same collection, function or truth predicate actually exists. Four tests will reveal their distinctive mathematics: Russell's collection, the category of all sets, Cantor's diagonal argument, and a universal satisfaction predicate.[1]

§1.1 Russell's predicate: exactly what may be collected?

The tempting comprehension formula

$$R = \{x : x \notin x\}$$

would imply $R \in R \iff R \notin R$. Each foundational system must reject some step in the construction.

In ZFC, one may *separate* elements from a set already given. If A is a set, the restricted collection

$$R_A = \{x \in A : x \notin x\}$$

is legitimate, and it satisfies the useful theorem

$$\boxed{R_A \notin A}.$$

For if $R_A \in A$, its own defining condition makes $R_A \in R_A$ equivalent to $R_A \notin R_A$. There is no paradox in constructing R_A; it simply lies outside the proposed container A. Consequently no universal *set* exists in ZFC.

NBG permits a Russell *class*

$$\mathcal{R} = \{x : x \text{ is a set and } x \notin x\}.$$

The diagonal argument shows that $\mathcal{R}$ must be a proper class, unable to occur as an element of another class. The universe V of all sets is similarly a proper class. KM gives the same answer to this problem; its stronger comprehension has not been invoked.

NF instead restricts the *syntax* of comprehension. A formula is stratified if variables can be assigned type indices so that $x \in y$ raises the type by one and equality leaves it unchanged. The formula $x = x$ is stratified, and so

$$\boxed{V = \{x : x = x\}}$$

is an actual universal set, with $V \in V$. But $x \notin x$ cannot be stratified: the type of x would have to exceed itself by one. Russell's collection is unavailable for a wholly different reason.

Thus ZFC restricts the domain of comprehension, class theories distinguish what can be a member, and NF restricts which predicates define sets.

[1] For class comprehension and truth predicates, see Victoria Gitman's expositions of second-order set theory. For stratified comprehension and Cantor's argument, see T. E. Forster, *Quine's New Foundations*, in the *Stanford Encyclopedia of Philosophy*, and E. P. Specker, *The Axiom of Choice in Quine's New Foundations* (1953).

§1.2 Large Categories and Expressive Power

For two fixed sets A, B, ZFC constructs the function set B^A without difficulty. It also proves the category laws by evaluating functions at elements:

$$((h \circ g) \circ f)(a) = h(g(f(a))) = (h \circ (g \circ f))(a).$$

But the objects of the category **Set** do not form a set. ZFC states the category laws uniformly about arbitrary sets and functions, using definable collections rather than pretending the entire category is small.

NBG can make the object collection and the collection of set-coded arrows into classes. Each $\mathrm{Hom}(A, B) = B^A$ remains a set, so **Set** is *locally small*. The familiar power-set assignment illustrates what becomes easier to package:

$$\mathcal{P}(f) : \mathcal{P}(A) \longrightarrow \mathcal{P}(B), \qquad S \longmapsto f[S].$$

Because $(g \circ f)[S] = g[f[S]]$, we obtain

$$\mathcal{P}(g \circ f) = \mathcal{P}(g) \circ \mathcal{P}(f).$$

ZFC proves this uniformly, NBG records the functor as class-sized data, and KM permits the same construction. No extra KM comprehension is needed here. Class notation can improve the language in which a theorem is stated without changing the set-level theorem.

The universal set of NF does not automatically settle such a category problem. Standard encodings of functions and ordered pairs must obey stratification; existence of a set of all objects alone does not prove that every intended collection of arrows has the desired set-coded operations.

§1.3 Cantor's Diagonal and the Universal Set

In ZFC, suppose $f : A \to \mathcal{P}(A)$ is surjective. Separation constructs

$$D = \{a \in A : a \notin f(a)\}.$$

Some $d \in A$ must satisfy $f(d) = D$, whence $d \in D \iff d \notin D$, impossible. NBG and KM prove the identical result for each set A.

In NF, however, every set is a subset of the universal set V, and every subset of V is a set. Therefore

$$\boxed{\mathcal{P}(V) = V},$$

so the identity map is a surjection from V onto its power set. Where has Cantor's proof gone? The predicate $a \notin f(a)$ is not stratified for an arbitrary set-coded map f: evaluating a function forces the input and output to have matching syntactic type levels, but the membership test requires the output one level higher. The diagonal set need not exist.

NF does retain a type-shifted Cantor comparison. The set of singletons

$$\{\{a\} : a \in A\}$$

must be distinguished from A itself: the usual singleton map $a \mapsto \{a\}$ need not exist as a set-coded function on an arbitrary NF set. Sets for which this map does exist are called *strongly Cantorian*. This is a concrete mathematical price of admitting a universal set.

§1.4 Uniform Truth in KM

Can one form a single class that says which first-order formulas are true in $(V, \in)$? A satisfaction class T would contain a formula code and an assignment to its free variables, and satisfy the Tarski clauses uniformly:

$$\begin{aligned} T(\ulcorner v_i \in v_j \urcorner, a) &\iff a_i \in a_j, \\ T(\ulcorner \neg\psi \urcorner, a) &\iff \neg T(\ulcorner \psi \urcorner, a), \\ T(\ulcorner \exists v_i\, \psi \urcorner, a) &\iff \exists x\, T(\ulcorner \psi \urcorner, a[i \mapsto x]). \end{aligned}$$

For each *fixed* formula, ZFC can define its truth relation by following its finite syntactic tree. But Tarski's undefinability theorem forbids a single first-order definable predicate handling *all* formulas of the universe at once.

NBG can quantify over classes, but its class comprehension permits only formulas without bound class quantifiers. It cannot prove that there is a full satisfaction class for V: a model containing only its definable classes would violate Tarski's theorem if it had one.

KM permits formulas defining classes to quantify over classes. This impredicative strength proves the existence of a uniform first-order satisfaction class. Its consequences are visible even in arithmetic. Such a truth class supports a formal soundness argument for ZFC: the axioms hold in V, the inference rules preserve truth, and a contradiction cannot be a theorem. Therefore

$$\boxed{\mathrm{KM} \vdash \mathrm{Con(ZFC)}.}$$

Assuming ZFC is consistent, Gödel's second incompleteness theorem means ZFC cannot prove this sentence. NBG is conservative over ZFC for set sentences, so it cannot prove it either. KM's additional comprehension is not simply a more convenient way to talk about the same classes.

§1.5 Choice: global convenience and the NF obstruction

ZFC's axiom of choice says that for each *set-indexed* family of nonempty sets $(A_i)_{i \in I}$, there exists a function c with $c(i) \in A_i$. It does not automatically provide a single class function choosing an element from *every* nonempty set in the universe.

A class theory can express a global choice function C:

$$\boxed{C(A) \in A \quad \text{for every nonempty set } A.}$$

One often writes GBC for Gödel–Bernays class theory augmented by Global Choice. This is a precise additional principle that can streamline uniform class-sized constructions, and it should not be silently identified with the ordinary axiom of choice on sets.

NF has a fundamentally different answer. Specker proved in 1953 that the ordinary axiom of choice fails in NF; indeed its universal set cannot be well-ordered. This is a theorem rather than an unsettled consistency issue. The related theory NFU allows urelements by weakening extensionality. Jensen's relative-consistency theorem shows that NFU can be consistent with Choice, so the contrast is not simply caused by permitting large universal sets.

The comparison now has real mathematical content. Russell's predicate tests the scope of comprehension; the category of all sets tests class-sized organization; Cantor's theorem tests which diagonal subsets may be formed; satisfaction classes test impredicative

comprehension; and Choice tests whether a family of selections can be assembled into a function. Each system answers these questions with different concrete constructions and theorems.